

- Sampling Theorem (Analog)
- $f_s \geq 2f_m \rightarrow f_m \rightarrow \text{max. Modulating Signal frequency}$
- $f_N = 2f_m \rightarrow \text{Nyquist rate of Sampling}$
- $T_N = \frac{1}{f_N} = \frac{1}{2f_m} \rightarrow \text{Nyquist Interval}$
- $f_s < 2f_m \rightarrow \text{Aliasing effect}$

Example 1

Find the Nyquist rate and interval for the signal $x(t) = 1 + \cos 10\pi t$ in Hz.

$$x(t) = 1 + \cos 10\pi t \quad \xrightarrow{\text{AC}} \quad \cos 2\pi f t$$

$\nwarrow$ $\swarrow$
DC AC

$$2\pi f = 10\pi$$

$$\boxed{f = 5 \text{ Hz}}$$

$$f_N = 2f_m \\ = 2 \times 5$$

$$\boxed{f_N = 10 \text{ Hz}}$$

$$T_N = \frac{1}{f_N}$$

$$T_N = \frac{1}{10} = \boxed{0.1 \text{ sec}}$$

Example 2

Find the Nyquist rate for the signal $x(t) = \sin 200\pi t - 2\cos 100\pi t$ in Hz.

$$x = x_1(t) + x_2(t)$$

$$x_1(t) = f_1 = 2\pi f = 200\pi$$
$$\boxed{f_1 = 100\text{ Hz}}$$

$$x_2(t) = f_2 = 2\pi f = 100\pi$$
$$\boxed{f_2 = 50\text{ Hz}}$$

$$f_{\max} = \max(f_1, f_2)$$

$$\boxed{f_{\max} = 100\text{ Hz}}$$

$$f_N = 2 f_{\max}$$

$$\boxed{f_N = 200\text{ Hz}}$$

Example 3

$$n/f_s$$

Consider an analog signal $x(t) = 5\cos 200\pi t$ (a) determine the minimum sampling rate to avoid aliasing (b) If sampling rate f_s is 400 Hz, what is discrete time signal after sampling?

$$f_N = 2 \times 100$$

$$f_N = 200 \text{ Hz}$$

$$f_m = 100 \text{ Hz}$$

$$f_s = 400 \text{ Hz}$$

Replace $t = n/f_s$ in $x(t)$

$$x(n) = 5\cos\left(200\pi \frac{n}{f_s}\right)$$

$$x(n) = 5\cos\left(\frac{200\pi \times n}{400}\right)$$

$$x(n) = 5\cos\left(\frac{n\pi}{2}\right)$$

$$x(n) = 5\cos(0.5\pi n)$$

Example 4

Find the Nyquist rate and Nyquist interval for the signal:

$$x(t) = 5\cos 1000\pi t \cdot \cos 4000\pi t$$

$$2\cos A \cdot \cos B = \cos(A+B) + \cos(A-B)$$

$$x(t) = 2.5\cos 5000\pi t + 2.5\cos 3000\pi t$$

$$f_1 = 2\pi f = 5000\pi$$

$$f_1 = 2500\text{ Hz}$$

$$f_2 = 2\pi f = 3000\pi$$

$$f_2 = 1500\text{ Hz}$$

$$f_{\max} = \max(f_1, f_2)$$

$$f_{\max} = 2500\text{ Hz}$$

$$f_N = 2f_{\max}$$

$$f_N = 5000\text{ Hz}$$

$$T_N = \frac{1}{f_N}$$

$$T_N = \frac{1}{5 \times 10^3}$$

$$T_N = 0.2\text{ msec}$$

Example 5

What is the value of minimum sampling frequency of a signal passed through an LPF with cut-off frequency 20 KHz?

- (a) 20 KHz (b) 30 KHz (c) ☒ 40 KHz (d) 50 KHz

$$2 \times 20 \text{ KHz} = 40 \text{ KHz}$$

Example 6

To satisfy the sampling theorem, a 100 Hz sine wave should be sampled at (a) 10 Hz (b) 100 Hz (c) ☒ 200 Hz (d) 50 Hz

$$2 \times 100 \text{ Hz} = 200 \text{ Hz}$$

Example 7

A system has a sampling rate of 50,000 samples per seconds. The maximum frequency of the signal it can acquire to reconstruct is:

- ✓ (a) 25 KHz (b) 50 KHz (c) 100 KHz (d) 10 KHz

$$f_s = 50000 = 50 \text{ KHz}$$

$$f_{\max} = 25 \text{ KHz}$$

Example 8

The spectrum of a band pass signal spans from 20 KHz to 30 KHz. The signal can be recovered ideally from the sampled values when the sampling rate is at least: (a) 20 KHz (b) 60 KHz (c) 50 KHz (d) 40 KHz

$$f_{\max} = 30 \text{ KHz}$$

Example 9

A signal of maximum frequency of 10 KHz is sampled at Nyquist rate. The time interval between two successive samples is

- (a) 50 μ s (b) 100 μ s (c) 500 μ s (d) 1000 μ s

$$\frac{1}{20 \times 10^3} = 50 \times 10^{-6} = 50 \mu\text{s}$$

Example 10

How many minimum number of samples are required to exactly describe the following signal? $x(t) = 10 \cos(6\pi t) + 4 \sin(8\pi t)$:

(a) 4 samples per second (b) 6 s/s (c) 8 s/s (d) 2 s/s

μs